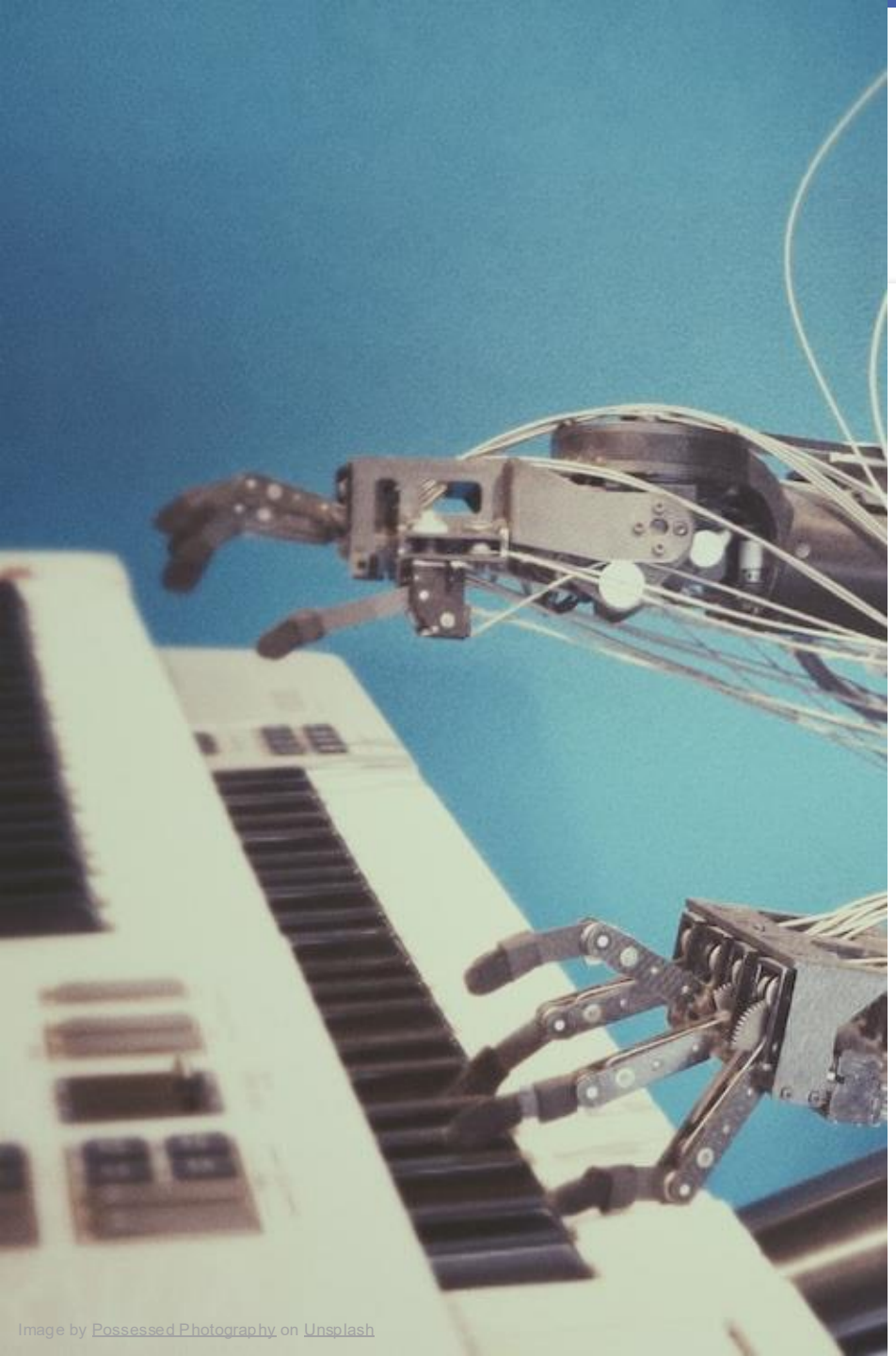


Applied Artificial Intelligence
Setup

Univ.-Prof. Dr-Ing. habil. Niklas Kühl
www.niklas.xyz

University Bayreuth

TUM School of Management



1. Goal for today
2. Let’s build our setup
3. Getting to know VS Code together



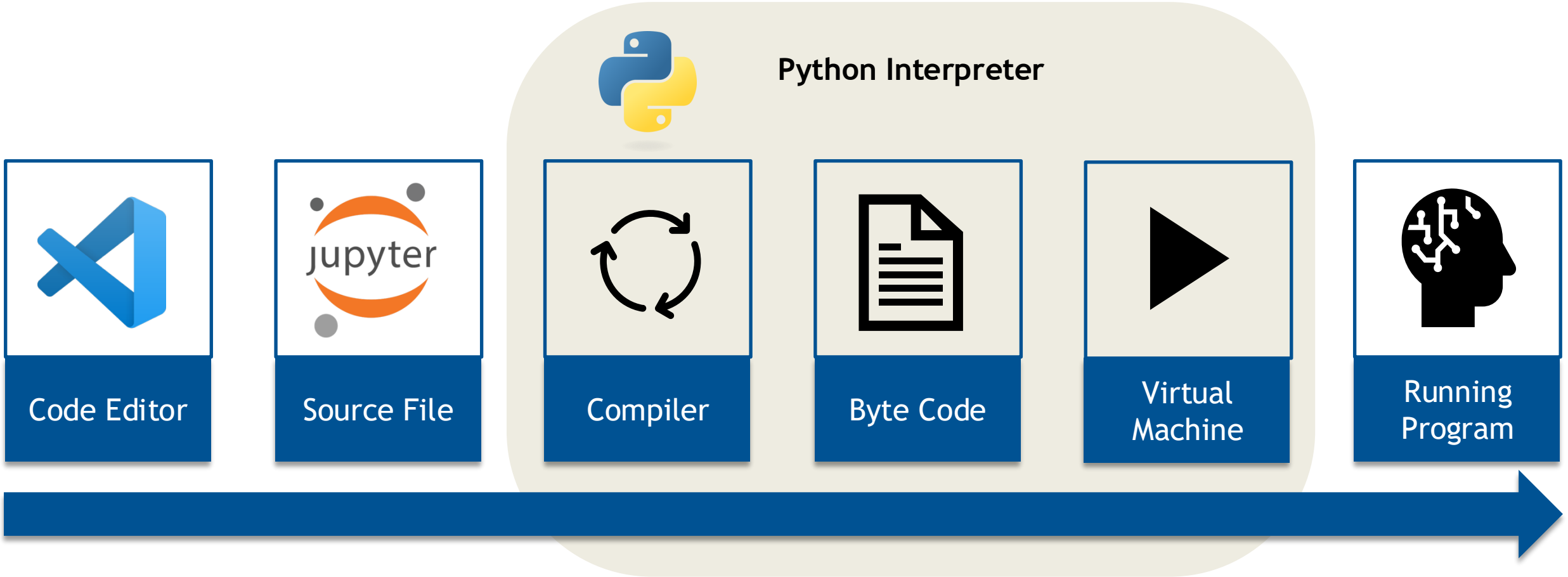
1. Goal for today
2. Let’s build our setup
3. Getting to know VS Code together

What do we want to do in our exercises?

Lecture	We learn about AI.
Exercise	We learn how to tell our computer to behave like an AI.

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How our computer executes code.



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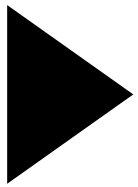
1. Goal for today
2. Let’s build our setup
3. Getting to know VS Code together

Let’s build our setup.









Python

Download
Installation

VS Code

Download
Installation
Extensions

Notebooks

Open folder in VS Code
Create an .ipynb file

Virtual Environment

Select Kernel
Run Notebook

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Already have a working Python interpreter?
Then you are good to go!

- There are many options to open, edit, and share code.
- If you already have a working setup you feel comfortable with, feel free to skip to section 3.

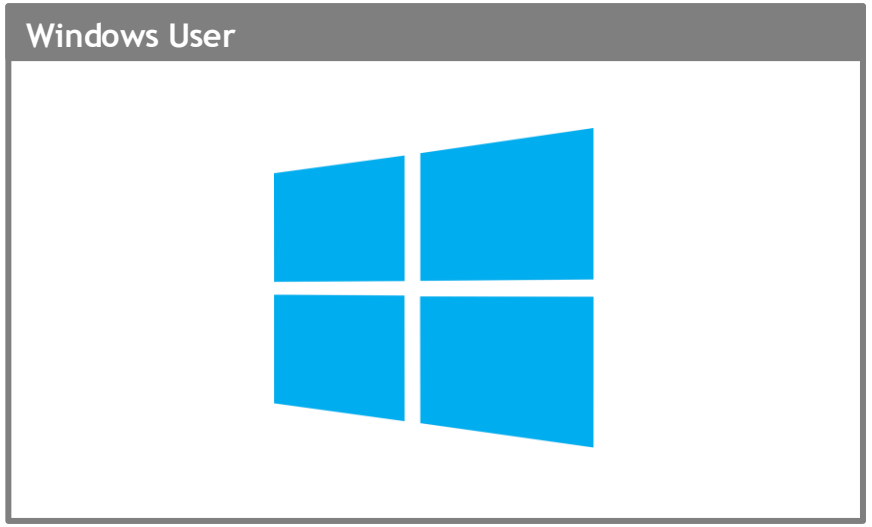
But first …

… it’s a decision…

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But first …

… are you a Windows or Mac user?





But first …

… are you a Windows or Mac user?

Windows User



Team Blue

Mac User

Team Orange



CC BY NC SA

But first …

Windows User



Team Blue

Mac User

Team Orange

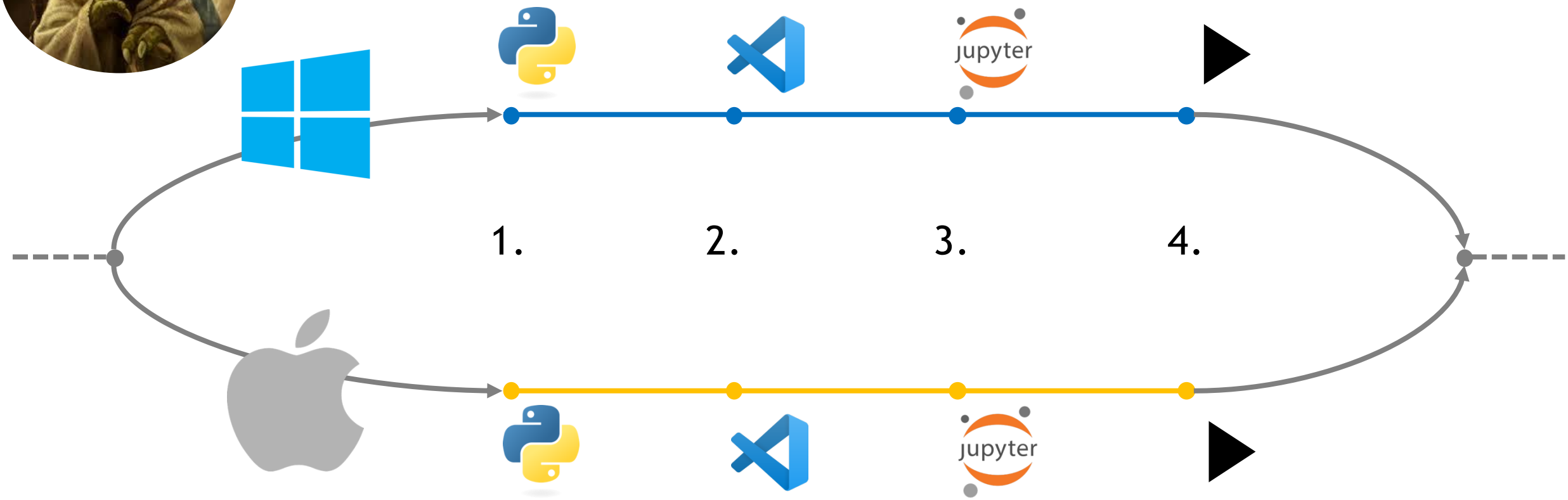


… please help each other anyway!

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Your path, you must follow.

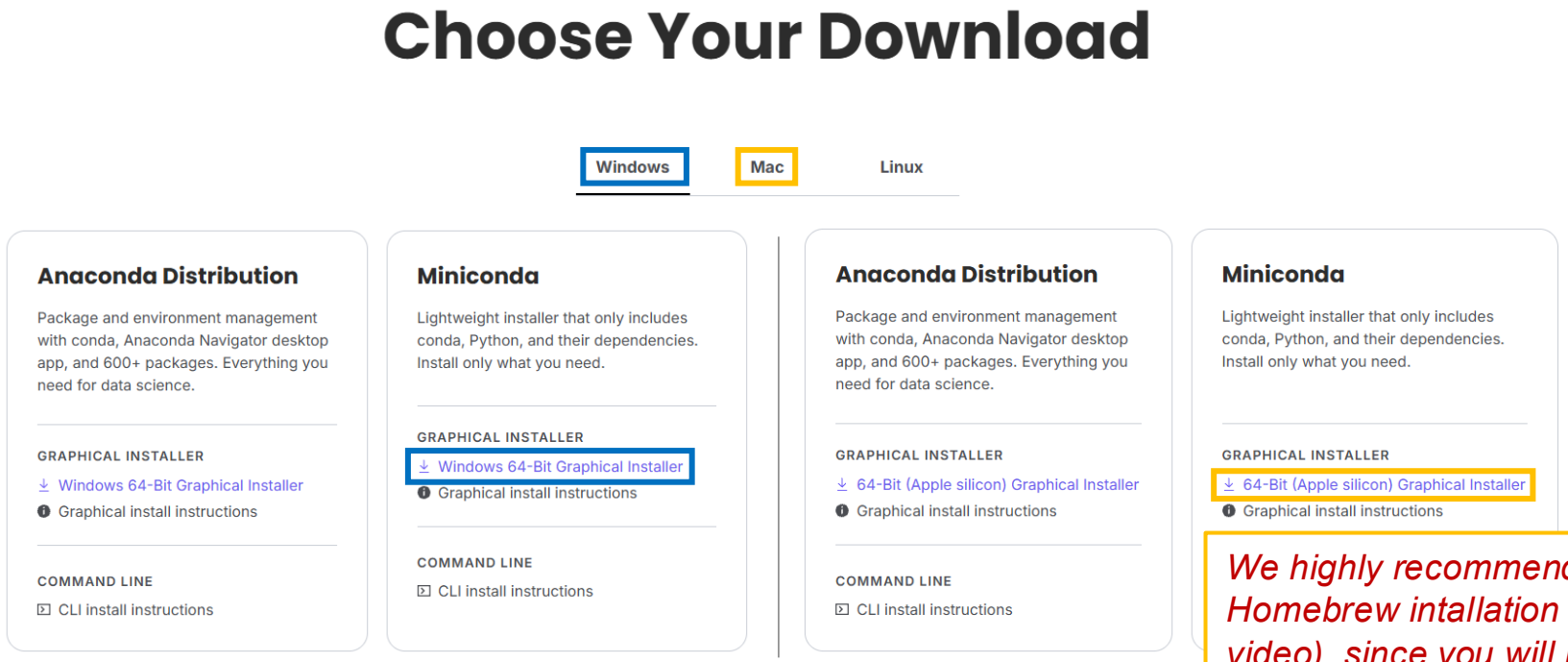




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Step 1: Creating our phyton environment
Step 1.1: Download miniconda installer

Visit the Anaconda website and download miniconda: https://www.anaconda.com/download/success



We highly recommend following the Homebrew intallation (see the youtube video), since you will need it later on for our git setup

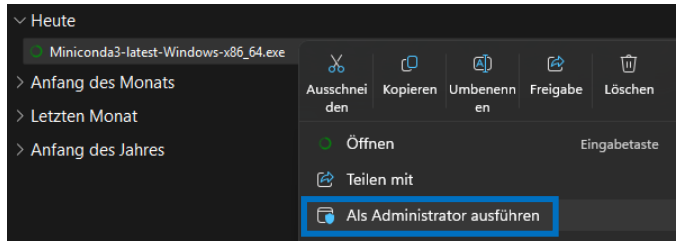
For Mac Users:

Already using ***Homebrew***? This YouTube video shows how to install Miniconda with Homebrew. Please follow the instructions only until 3:26 min: https://www.youtube.com/watch?v=WdXdl0C0jfE. Afterwards, go to **Step 2**.

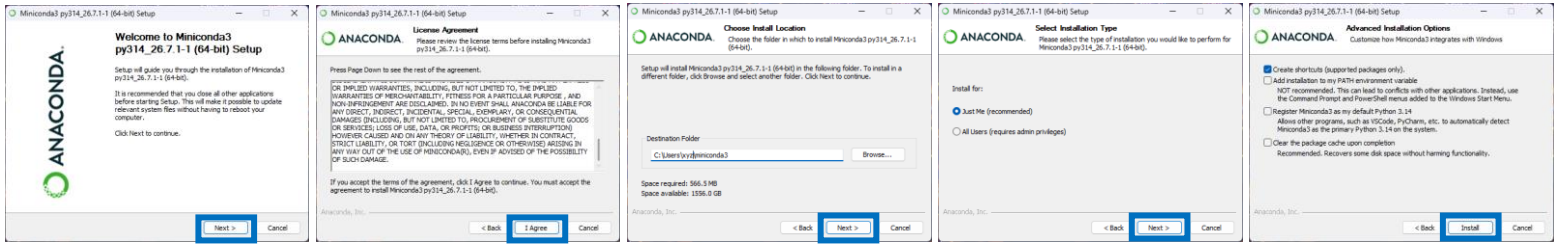
Prof. Dr. Niklas Kühl

Step 1: Creating our Phyton environment
Step 1.2: Install miniconda **(Windows)**

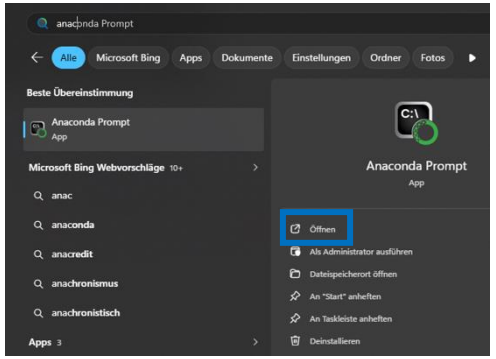
(i) Open the „.exe“-file as administrator



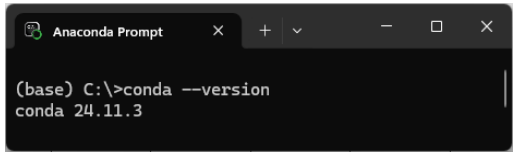
(ii) Click through the installation wizard



(iii) Search for “anaconda” in the windows-search and open it



(iv) Type “conda --version“ to verify the installation. It should display something like “conda xx.xx.x”

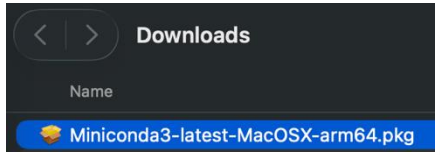


**Please note: While the interface is shown in German, the layout remains exactly the same.*

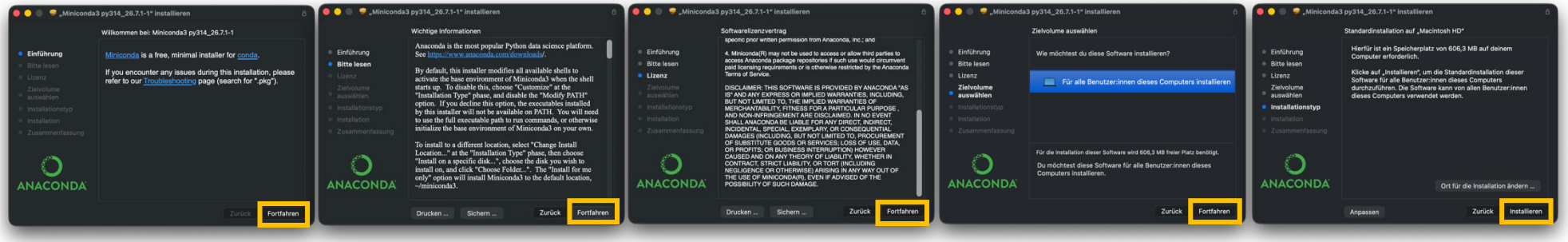
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Step 1: Creating our Phyton environment
Step 1.2: Install miniconda (MacOS)

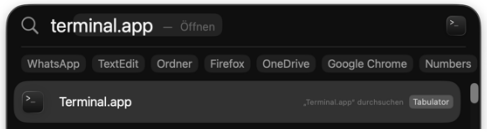
(i) Double-click the „.pkg“-file



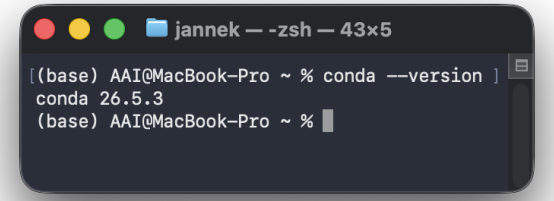
(ii) Click through the installation wizard



(iii) Open spotlight, search for “terminal” and open it



(iv) Type “conda --version“ to verify the installation.
It should display something like “conda xx.xx.x”



**Please note: While the interface is shown in German, the layout remains exactly the same.*

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Step 1: Creating our Phyton environment
Step 1.3: Finally let's create our environment

This step is identical for Windows and MacOS users: Type in or copy the following lines of code and execute them line by line.

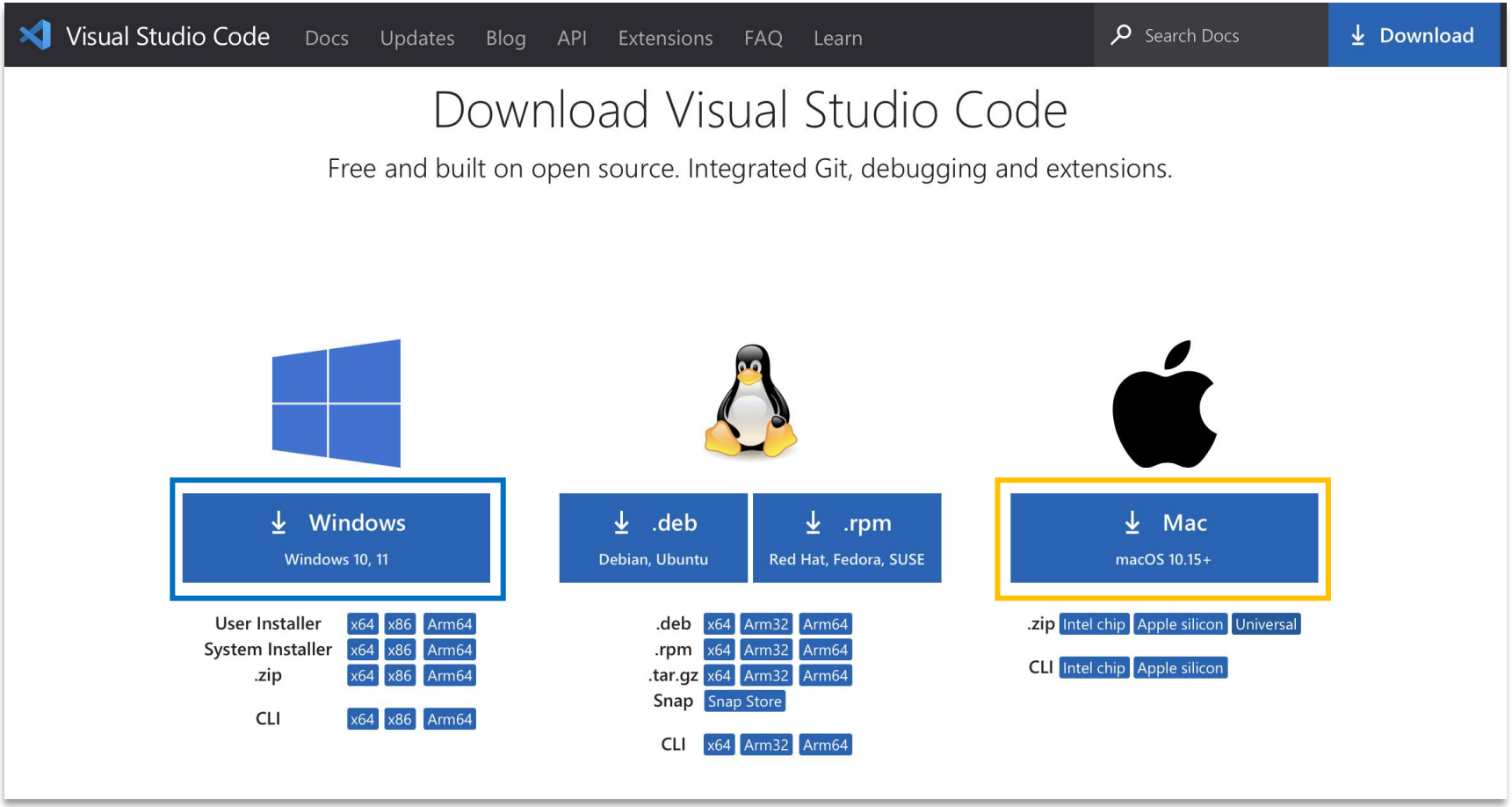
```
line 1  conda create -n AAI python=3.14 pip
line 2  conda activate AAI
line 3  pip install pandas matplotlib scikit-learn seaborn numpy scipy jupyter ipykernel
```

Congratulations! You just set up your very first Python environment, AAI.

**Please note: You should already be in your "Anaconda Prompt" or "Terminal", if not refer to step 1.2 (iii)!*

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Step 2: Initiate the IDE
Step 2.1: Download and install VS Code



https://code.visualstudio.com/Download

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Step 2: Initiate the IDE
Step 2.2: Open VS Code

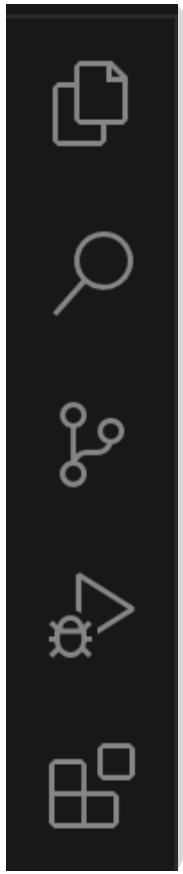
EXPLORER

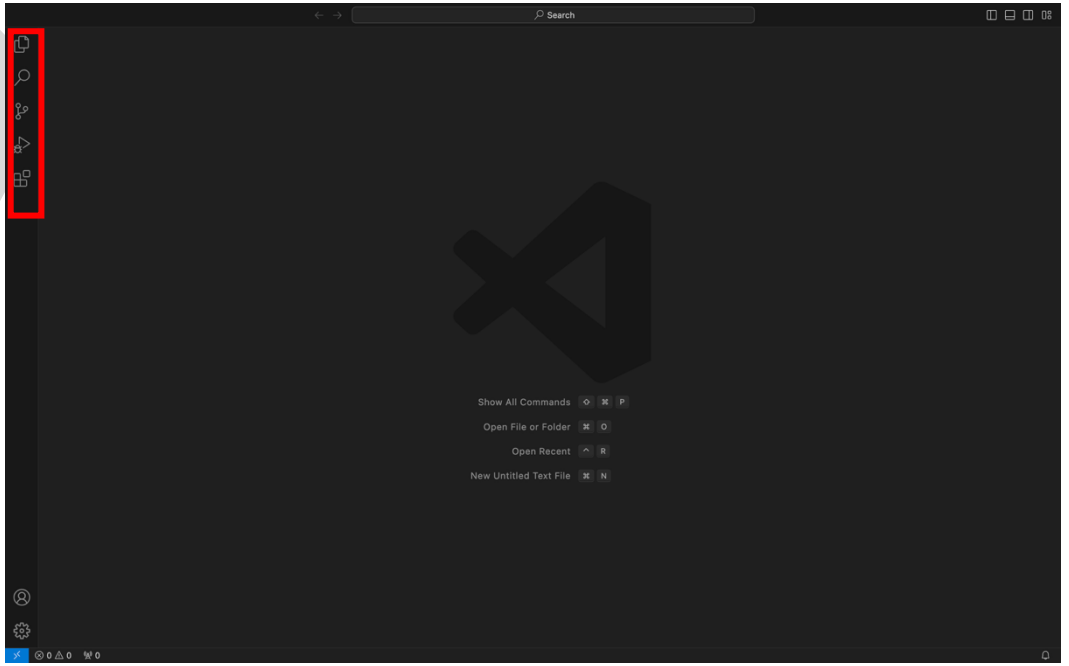
SEARCH

SOURCE CONTROL

RUN AND DEBUG

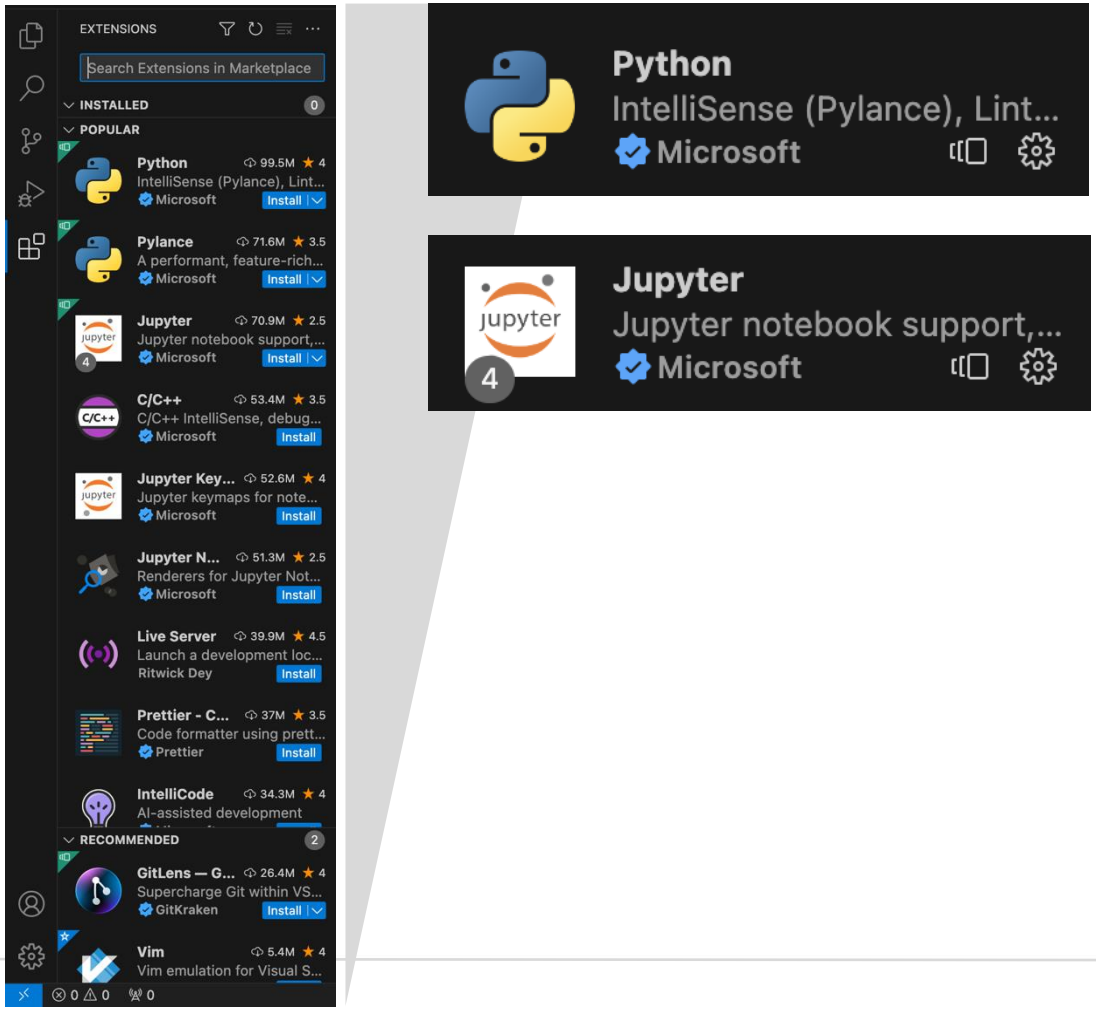
EXTENSIONS





Prof. Dr. Niklas Kühl

Step 2: Initiate the IDE
Step 2.3: Install VS Code Extensions



1. Python

2. Jupyter

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Step 2: Initiate the IDE
Step 2.3: Install VS Code Extensions

Windows User

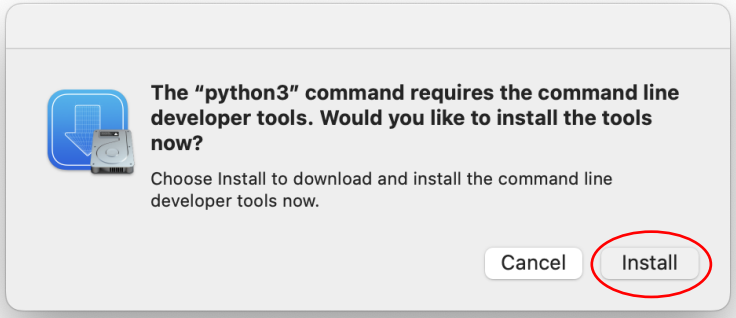
Install extensions

Install developer tools (shows up automatically)!

Mac User

Install extensions

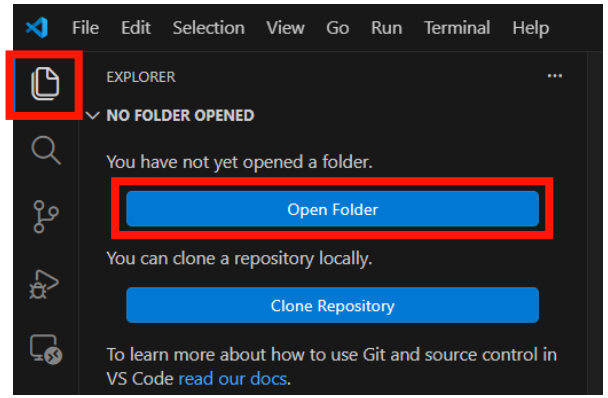
Install developer tools (shows up automatically)!

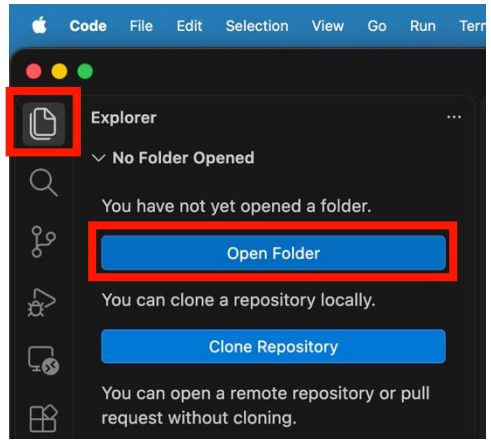


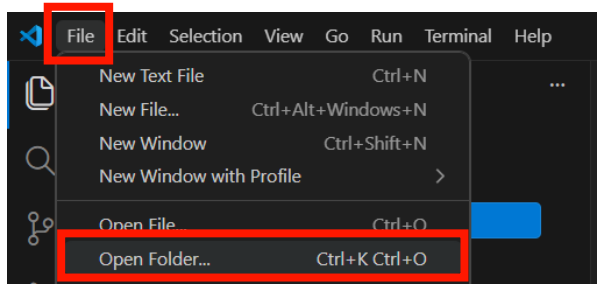
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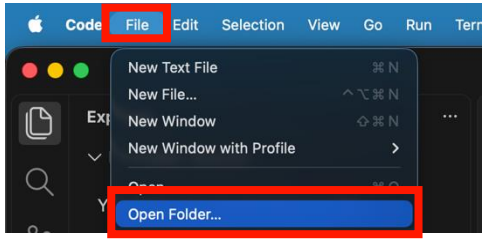
Step 3: Try out Python in VS Code
Step 3.1: Open a new folder and create your first notebook

(i) Go to the “explorer” tab and click “Open Folder” or click on “File” and “Open Folder…”.
The Layout is identical on Windows and MacOS.

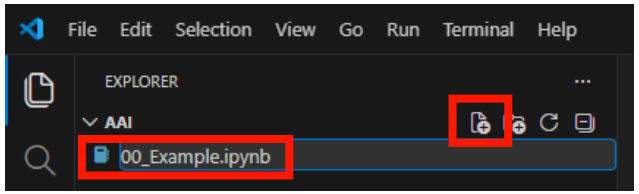








(ii) Create a new file with the ending “.ipynb”
The Layout is identical on Windows and MacOS.

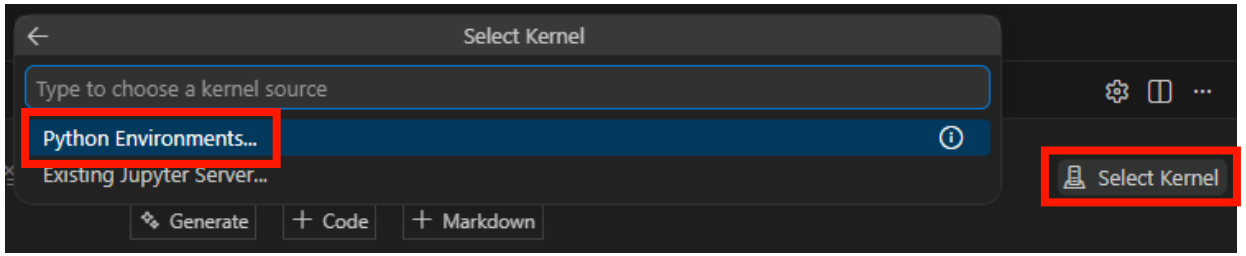


**Please note: The next steps are identical on Windows and MacOS*

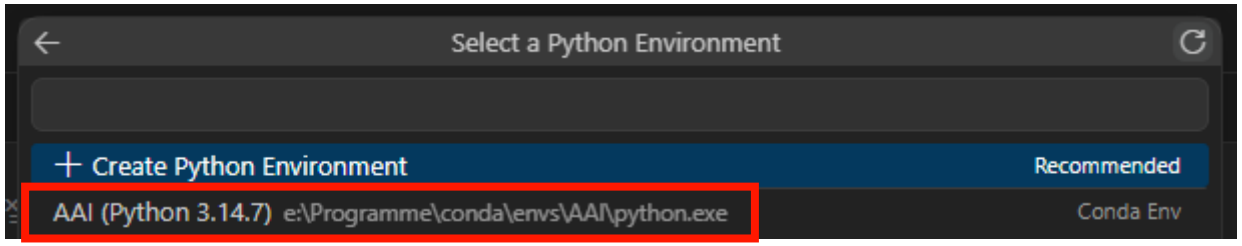
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Step 3: Try out Python in VS Code
Step 3.2: Select our AAI environment

(i) In the created .ipynb file click on “Select Kernel” and “Python Environments…”



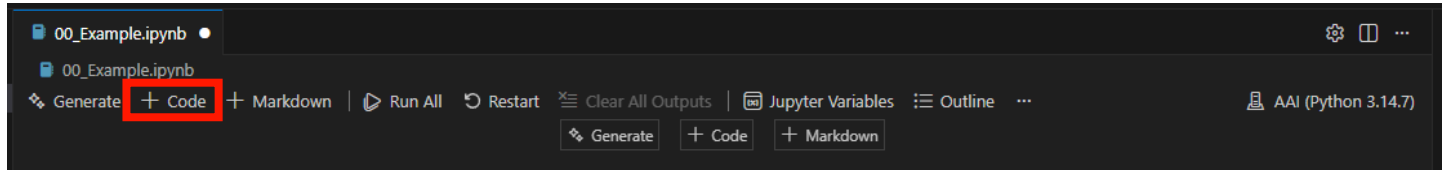
(ii) Now select our AAI environment



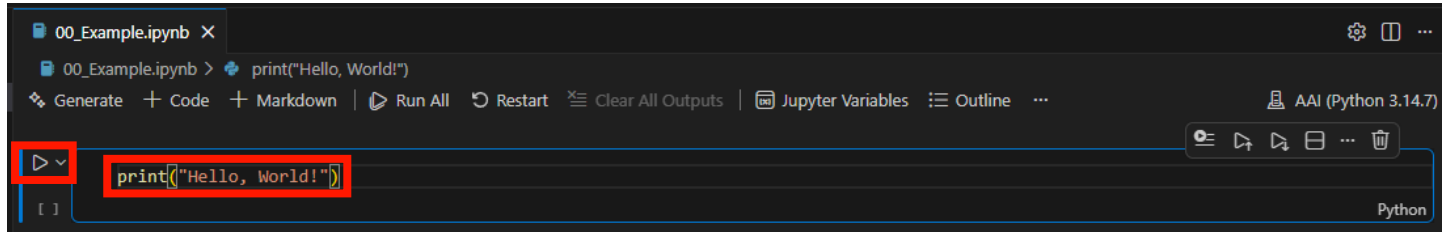
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Step 3: Try out Python in VS Code
Step 3.3: Hello World example

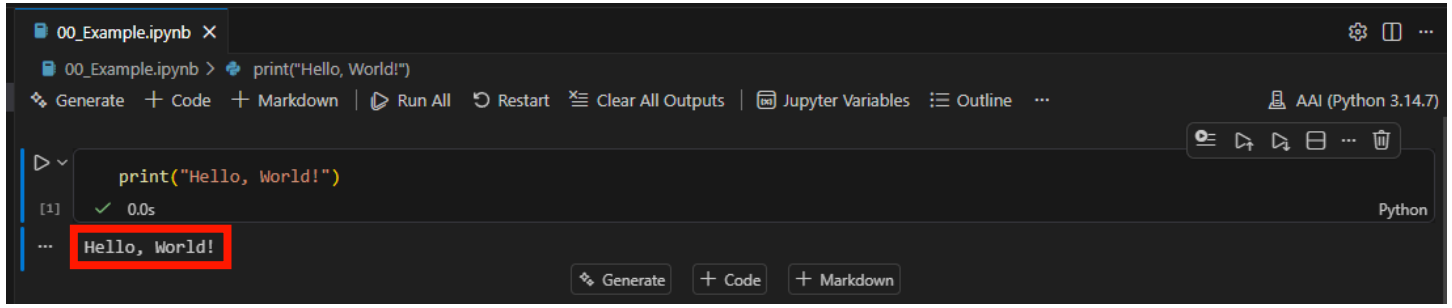
(i) Create a new “Code”-Cell



(ii) Type in the displayed code and click the run button



(iii) Look at the results



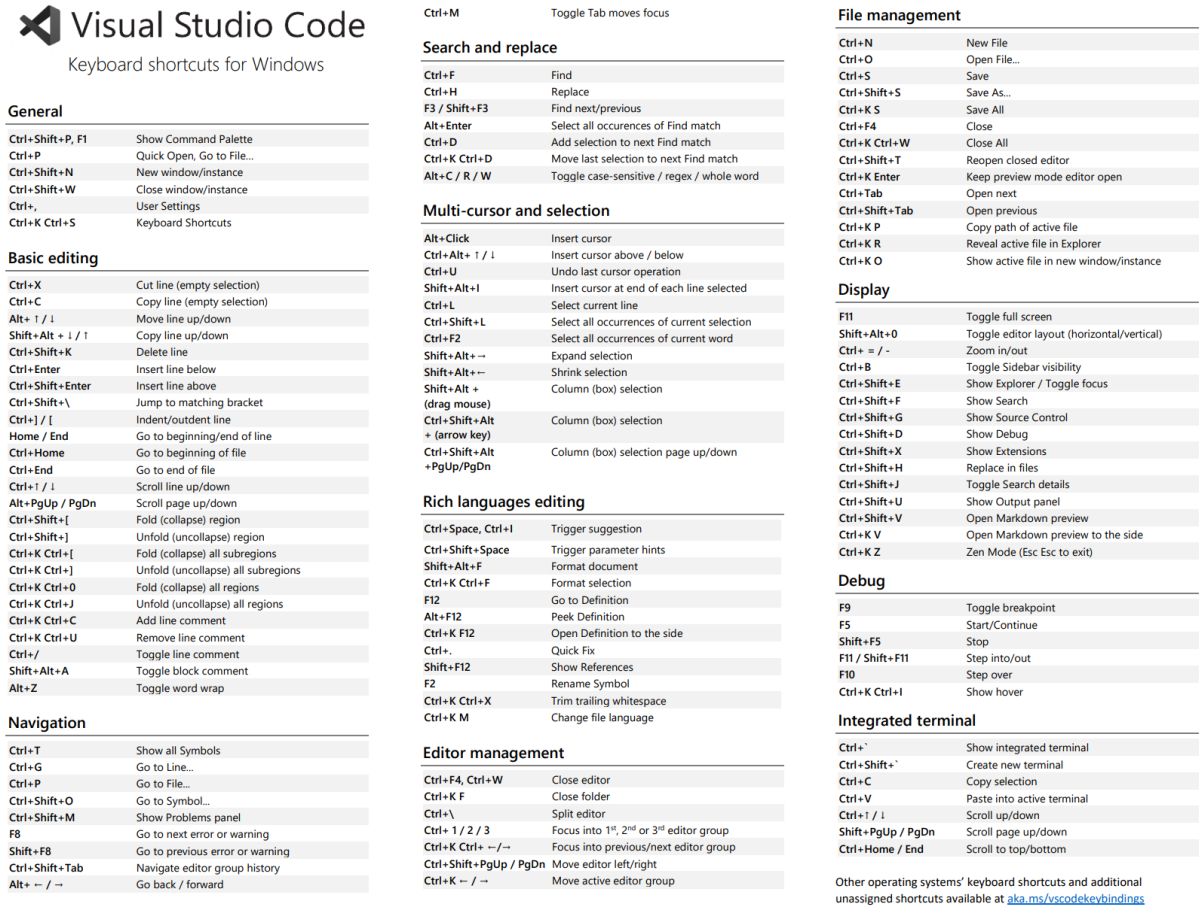
Prof. Dr. Niklas Kühl



1. Goal for today
2. Let’s build our setup
3. Getting to know VS Code together

Some useful VS Code shortcuts (Windows)

https://code.visualstudio.com/shortcuts/keyboard-shortcuts-windows.pdf



Visual Studio Code

Keyboard shortcuts for Windows

General

Ctrl+Shift+P, F1	Show Command Palette
Ctrl+P	Quick Open, Go to File...
Ctrl+Shift+N	New window/instance
Ctrl+Shift+W	Close window/instance
Ctrl+,	User Settings
Ctrl+K Ctrl+S	Keyboard Shortcuts

Basic editing

Ctrl+X	Cut line (empty selection)
Ctrl+C	Copy line (empty selection)
Alt+ ↑ / ↓	Move line up/down
Shift+Alt + ↓ / ↑	Copy line up/down
Ctrl+Shift+K	Delete line
Ctrl+Enter	Insert line below
Ctrl+Shift+Enter	Insert line above
Ctrl+Shift+\	Jump to matching bracket
Ctrl+] / [	Indent/outdent line
Home / End	Go to beginning/end of line
Ctrl+Home	Go to beginning of file
Ctrl+End	Go to end of file
Ctrl+↑ / ↓	Scroll line up/down
Alt+PgUp / PgDn	Scroll page up/down
Ctrl+Shift+[	Fold (collapse) region
Ctrl+Shift+]	Unfold (uncollapse) region
Ctrl+K Ctrl+[	Fold (collapse) all subregions
Ctrl+K Ctrl+]	Unfold (uncollapse) all subregions
Ctrl+K Ctrl+0	Fold (collapse) all regions
Ctrl+K Ctrl+J	Unfold (uncollapse) all regions
Ctrl+K Ctrl+C	Add line comment
Ctrl+K Ctrl+U	Remove line comment
Ctrl+/	Toggle line comment
Shift+Alt+A	Toggle block comment
Alt+Z	Toggle word wrap

Navigation

Ctrl+T	Show all Symbols
Ctrl+G	Go to Line...
Ctrl+P	Go to File...
Ctrl+Shift+O	Go to Symbol...
Ctrl+Shift+M	Show Problems panel
F8	Go to next error or warning
Shift+F8	Go to previous error or warning
Ctrl+Shift+Tab	Navigate editor group history
Alt+ ← / →	Go back / forward
Ctrl+M	Toggle Tab moves focus

Search and replace

Ctrl+F	Find
Ctrl+H	Replace
F3 / Shift+F3	Find next/previous
Alt+Enter	Select all occurences of Find match
Ctrl+D	Add selection to next Find match
Ctrl+K Ctrl+D	Move last selection to next Find match
Alt+C / R / W	Toggle case-sensitive / regex / whole word

Multi-cursor and selection

Alt+Click	Insert cursor
Ctrl+Alt+ ↑ / ↓	Insert cursor above / below
Ctrl+U	Undo last cursor operation
Shift+Alt+I	Insert cursor at end of each line selected
Ctrl+L	Select current line
Ctrl+Shift+L	Select all occurrences of current selection
Ctrl+F2	Select all occurrences of current word
Shift+Alt+→	Expand selection
Shift+Alt+←	Shrink selection
Shift+Alt + (drag mouse)	Column (box) selection
Ctrl+Shift+Alt + (arrow key)	Column (box) selection
Ctrl+Shift+Alt +PgUp/PgDn	Column (box) selection page up/down

Rich languages editing

Ctrl+Space, Ctrl+I	Trigger suggestion
Ctrl+Shift+Space	Trigger parameter hints
Shift+Alt+F	Format document
Ctrl+K Ctrl+F	Format selection
F12	Go to Definition
Alt+F12	Peek Definition
Ctrl+K F12	Open Definition to the side
Ctrl+.	Quick Fix
Shift+F12	Show References
F2	Rename Symbol
Ctrl+K Ctrl+X	Trim trailing whitespace
Ctrl+K M	Change file language

Editor management

Ctrl+F4, Ctrl+W	Close editor
Ctrl+K F	Close folder
Ctrl+\	Split editor
Ctrl+ 1 / 2 / 3	Focus into 1st, 2nd or 3rd editor group
Ctrl+K Ctrl+ ←/→	Focus into previous/next editor group
Ctrl+Shift+PgUp / PgDn	Move editor left/right
Ctrl+K ← / →	Move active editor group

File management

Ctrl+N	New File
Ctrl+O	Open File...
Ctrl+S	Save
Ctrl+Shift+S	Save As...
Ctrl+K S	Save All
Ctrl+F4	Close
Ctrl+K Ctrl+W	Close All
Ctrl+Shift+T	Reopen closed editor
Ctrl+K Enter	Keep preview mode editor open
Ctrl+Tab	Open next
Ctrl+Shift+Tab	Open previous
Ctrl+K P	Copy path of active file
Ctrl+K R	Reveal active file in Explorer
Ctrl+K O	Show active file in new window/instance

Display

F11	Toggle full screen
Shift+Alt+0	Toggle editor layout (horizontal/vertical)
Ctrl+ = / -	Zoom in/out
Ctrl+B	Toggle Sidebar visibility
Ctrl+Shift+E	Show Explorer / Toggle focus
Ctrl+Shift+F	Show Search
Ctrl+Shift+G	Show Source Control
Ctrl+Shift+D	Show Debug
Ctrl+Shift+X	Show Extensions
Ctrl+Shift+H	Replace in files
Ctrl+Shift+J	Toggle Search details
Ctrl+Shift+U	Show Output panel
Ctrl+Shift+V	Open Markdown preview
Ctrl+K V	Open Markdown preview to the side
Ctrl+K Z	Zen Mode (Esc Esc to exit)

Debug

F9	Toggle breakpoint
F5	Start/Continue
Shift+F5	Stop
F11 / Shift+F11	Step into/out
F10	Step over
Ctrl+K Ctrl+I	Show hover

Integrated terminal

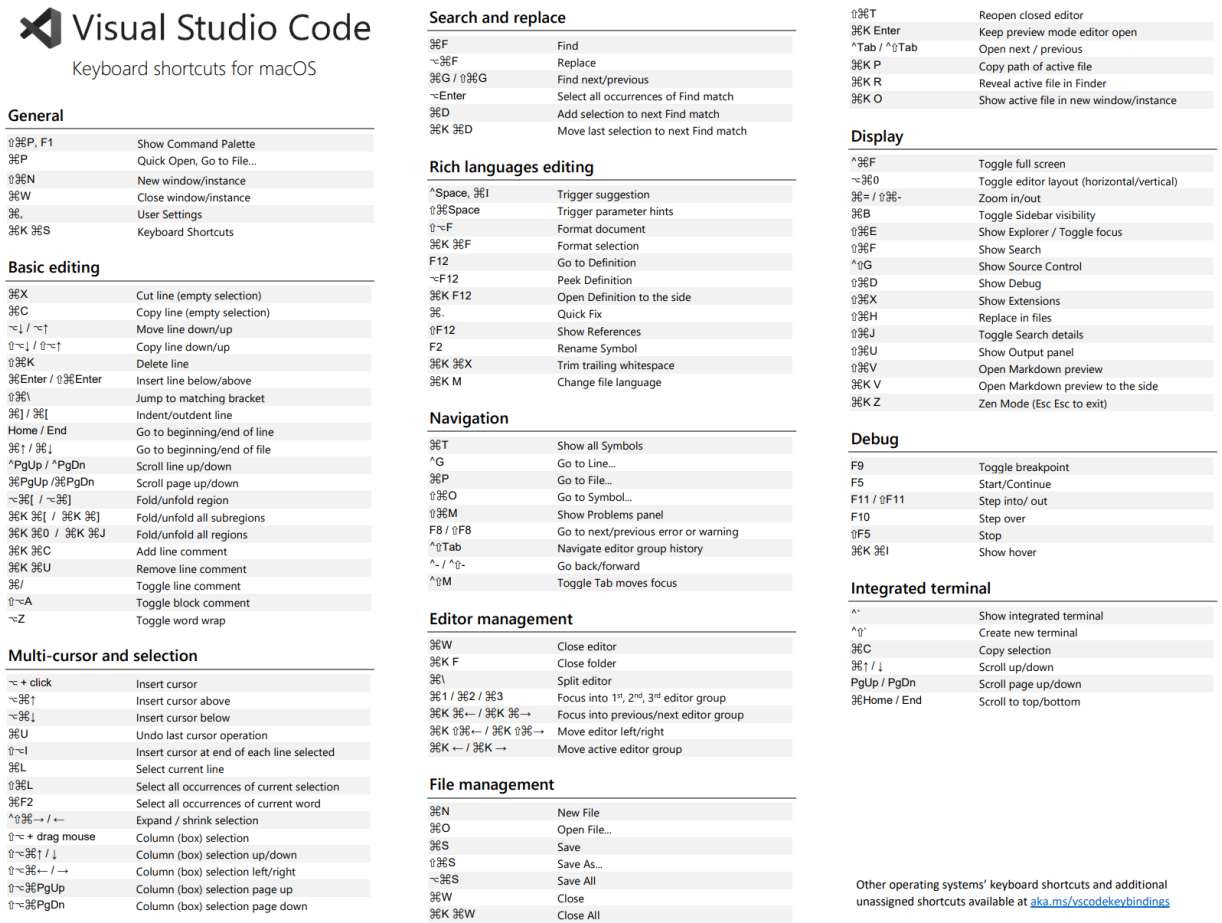
Ctrl+`	Show integrated terminal
Ctrl+Shift+`	Create new terminal
Ctrl+C	Copy selection
Ctrl+V	Paste into active terminal
Ctrl+↑ / ↓	Scroll up/down
Shift+PgUp / PgDn	Scroll page up/down
Ctrl+Home / End	Scroll to top/bottom

Other operating systems' keyboard shortcuts and additional unassigned shortcuts available at aka.ms/vscodekeybindings

Prof. Dr. Niklas Kühl

Some useful VS Code shortcuts (MacOS)

https://code.visualstudio.com/shortcuts/keyboard-shortcuts-macos.pdf



Visual Studio Code

Keyboard shortcuts for macOS

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⌘P	Quick Open, Go to File...
⇧⌘N	New window/instance
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⌘,	User Settings
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Basic editing

⌘X	Cut line (empty selection)
⌘C	Copy line (empty selection)
⌥↓ / ⌥↑	Move line down/up
⇧⌥↓ / ⇧⌥↑	Copy line down/up
⇧⌘K	Delete line
⌘Enter / ⇧⌘Enter	Insert line below/above
⇧⌘\	Jump to matching bracket
⌘] / ⌘[	Indent/outdent line
Home / End	Go to beginning/end of line
⌘↑ / ⌘↓	Go to beginning/end of file
^PgUp / ^PgDn	Scroll line up/down
⌘PgUp /⌘PgDn	Scroll page up/down
⌥⌘[/ ⌥⌘]	Fold/unfold region
⌘K ⌘[/ ⌘K ⌘]	Fold/unfold all subregions
⌘K ⌘0 / ⌘K ⌘J	Fold/unfold all regions
⌘K ⌘C	Add line comment
⌘K ⌘U	Remove line comment
⌘/	Toggle line comment
⇧⌥A	Toggle block comment
⌥Z	Toggle word wrap

Multi-cursor and selection

⌥ + click	Insert cursor
⌥⌘↑	Insert cursor above
⌥⌘↓	Insert cursor below
⌘U	Undo last cursor operation
⇧⌥I	Insert cursor at end of each line selected
⌘L	Select current line
⇧⌘L	Select all occurrences of current selection
⌘F2	Select all occurrences of current word
^⇧⌘→ / ←	Expand / shrink selection
⇧⌥ + drag mouse	Column (box) selection
⇧⌥⌘↑ / ↓	Column (box) selection up/down
⇧⌥⌘← / →	Column (box) selection left/right
⇧⌥⌘PgUp	Column (box) selection page up
⇧⌥⌘PgDn	Column (box) selection page down

Search and replace

⌘F	Find
⌥⌘F	Replace
⌘G / ⇧⌘G	Find next/previous
⌥Enter	Select all occurrences of Find match
⌘D	Add selection to next Find match
⌘K ⌘D	Move last selection to next Find match

Rich languages editing

^Space, ⌘I	Trigger suggestion
⇧⌘Space	Trigger parameter hints
⇧⌥F	Format document
⌘K ⌘F	Format selection
F12	Go to Definition
⌥F12	Peek Definition
⌘K F12	Open Definition to the side
⌘.	Quick Fix
⇧F12	Show References
F2	Rename Symbol
⌘K ⌘X	Trim trailing whitespace
⌘K M	Change file language

Navigation

⌘T	Show all Symbols
^G	Go to Line...
⌘P	Go to File...
⇧⌘O	Go to Symbol...
⇧⌘M	Show Problems panel
F8 / ⇧F8	Go to next/previous error or warning
^⇧Tab	Navigate editor group history
^- / ^⇧-	Go back/forward
^⇧M	Toggle Tab moves focus

Editor management

⌘W	Close editor
⌘K F	Close folder
⌘\	Split editor
⌘1 / ⌘2 / ⌘3	Focus into 1st, 2nd, 3rd editor group
⌘K ⌘← / ⌘K ⌘→	Focus into previous/next editor group
⌘K ⇧⌘← / ⌘K ⇧⌘→	Move editor left/right
⌘K ← / ⌘K →	Move active editor group

File management

⌘N	New File
⌘O	Open File...
⌘S	Save
⇧⌘S	Save As...
⌥⌘S	Save All
⌘W	Close
⌘K ⌘W	Close All
⇧⌘T	Reopen closed editor
⌘K Enter	Keep preview mode editor open
^Tab / ^⇧Tab	Open next / previous
⌘K P	Copy path of active file
⌘K R	Reveal active file in Finder
⌘K O	Show active file in new window/instance

Display

^⌘F	Toggle full screen
⌥⌘0	Toggle editor layout (horizontal/vertical)
⌘= / ⇧⌘-	Zoom in/out
⌘B	Toggle Sidebar visibility
⇧⌘E	Show Explorer / Toggle focus
⇧⌘F	Show Search
^⇧G	Show Source Control
⇧⌘D	Show Debug
⇧⌘X	Show Extensions
⇧⌘H	Replace in files
⇧⌘J	Toggle Search details
⇧⌘U	Show Output panel
⇧⌘V	Open Markdown preview
⌘K V	Open Markdown preview to the side
⌘K Z	Zen Mode (Esc Esc to exit)

Debug

F9	Toggle breakpoint
F5	Start/Continue
F11 / ⇧F11	Step into/ out
F10	Step over
⇧F5	Stop
⌘K ⌘I	Show hover

Integrated terminal

^`	Show integrated terminal
^⇧`	Create new terminal
⌘C	Copy selection
⌘↑ / ↓	Scroll up/down
PgUp / PgDn	Scroll page up/down
⌘Home / End	Scroll to top/bottom

Other operating systems' keyboard shortcuts and additional unassigned shortcuts available at aka.ms/vscodekeybindings

Prof. Dr. Niklas Kühl

Until next week

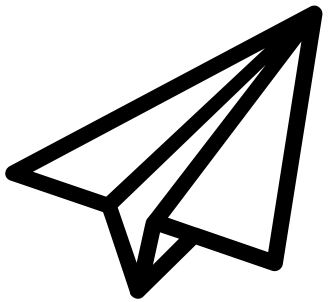
Make sure your setup works!

- Install Python
- Install VS Code
- Install Extensions
- Test your setup



Problems?

Reach out in the moodle/elearning forum.



Prof. Dr. Niklas Kühl